

LectureLens

An offline, on-device AI classroom companion for Snapdragon-powered HP PCs

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1. Summary

LectureLens is a Windows on ARM desktop application that listens to a live lecture, reads the whiteboard or slides through the webcam, and turns both into structured bilingual notes, flashcards, a short quiz and a searchable "ask your lecture" assistant. Every AI model runs locally on the Snapdragon Hexagon NPU, so the app works fully offline, keeps student data private and preserves battery life through a full day of classes.

2. Problem

- **Language gap:** many first-generation and regional-medium students in India find it hard to follow fast English-medium lectures.
- **Notes vs. attention:** students must choose between writing notes and understanding the teacher.
- **Connectivity:** campus Wi-Fi and rural internet make cloud AI tools slow or unusable.
- **Privacy and cost:** uploading class recordings to cloud services raises privacy concerns and usually needs paid subscriptions.

3. Solution and key features

- **Listen:** real-time speech-to-text of the teacher using Whisper on the NPU, shown as live captions.
- **See:** the webcam captures the board or slides; an object detector finds the board region and OCR extracts the text and formulas.
- **Understand:** a small on-device language model merges transcript and board text into structured notes, key terms, a summary, flashcards and a 5-question quiz.
- **Translate:** one click converts notes into Tamil, Hindi or other Indian languages.
- **Ask:** after class, students ask questions about the lecture; local semantic search finds the relevant part and the LLM answers from it.

4. AI models (Qualcomm AI Hub and open source)

Task	Model	Target
Speech-to-text	Whisper (small/base) – Qualcomm AI Hub	NPU
Board region detection	YOLO detector – Qualcomm AI Hub	NPU
Board/slide text	TrOCR / EasyOCR – Qualcomm AI Hub / open source	NPU
Notes, quiz, Q&A	Llama 3.2 3B or Phi-3.5-mini (INT4) – Qualcomm AI Hub	NPU

Translation	IndicTrans2 (AI4Bharat, open source)	NPU/CPU
Semantic search	MiniLM sentence embeddings (open source)	NPU/CPU

Models are quantized and compiled for Snapdragon X-series devices through Qualcomm AI Hub and executed with ONNX Runtime using the QNN Execution Provider, with CPU fallback for unsupported operators.

5. Technical implementation

- **Engine:** modular Python package with separate ASR, vision, notes, translation and Q&A modules behind a common interface, so each model can be swapped or upgraded independently.
- **Runtime:** ONNX Runtime (QNN EP) targeting the Hexagon NPU; audio is processed in short rolling windows for low-latency captions.
- **UI:** lightweight desktop interface with a live view (captions + board text) and a lecture library (notes, translation, flashcards, quiz, Q&A).
- **Storage:** each class is saved as a local "lecture pack" (transcript, notes, images, embeddings) in SQLite and plain files.

6. Why Snapdragon-powered HP PCs

- The dedicated NPU runs speech, vision and language models continuously while the CPU and GPU stay free for the student's other work.
- NPU inference is much more power-efficient than CPU/GPU, enabling a full day of classes on battery.
- On-device processing means lecture audio, images and notes never leave the laptop.
- Fully offline operation suits campuses, hostels, travel and rural colleges.

7. Deployment and accessibility

- One-click Windows ARM64 installer with models bundled; no account, Python setup or internet needed.
- Interface and notes in English, Tamil and Hindi, extendable to more Indian languages.
- Large-text and high-contrast modes, keyboard navigation and text-to-speech of notes for visually impaired students; live captions for students with hearing difficulties.
- Open-source code on GitHub so colleges and developers can extend it.

8. Impact

LectureLens helps regional-medium and first-generation students keep up in English-taught classes, frees attention from note-taking to understanding, and turns every lecture into revision material automatically, at zero running cost and with full privacy.

9. Roadmap

- **Phase 1:** core prototype – live transcript, notes and quiz on the NPU.
- **Phase 2:** board OCR, Tamil/Hindi translation and lecture Q&A.
- **Phase 3:** accessibility features, installer and a pilot with students at VIT.
- **Phase 4:** more languages, teacher dashboard and rollout to other colleges.